

Active Directory Domain Setup and Network Services Configuration for “musicstream.local” (Windows Server 2019)

1. Create a New Active Directory Domain and User Accounts

Step 1: Install Active Directory Domain Services (AD DS)

1. Open **Server Manager**.
2. Click **Add Roles and Features**.
3. Select **Role-based or feature-based installation**.
4. Choose your server.
5. Install:
 - **Active Directory Domain Services**
6. Click **Install**.

Step 2: Promote Server to Domain Controller

1. After installation, click **Promote this server to a domain controller**.
2. Select:
 - **Add a new forest**
3. Enter the domain name:
musicstream.local
4. Set the **DSRM password**.
5. Complete the wizard.
6. Restart the server.

Step 3: Create 5 User Accounts

Open:

Active Directory Users and Computers

Go to:

musicstream.local → Users → New → User

Create the following users:

Department Username

Admin	admin_user
Marketing	marketing_user
Tech	tech_user
Support	support_user
HR	hr_user

Set passwords for the accounts and enable the accounts.

2. Configure DNS and DHCP Services

Step 1: DNS Configuration

DNS is automatically installed with AD DS, but verify the configuration.

1. Open **Server Manager**.
2. Go to:

Tools → DNS

3. Check the **Forward Lookup Zones**.
4. Verify that the following zone exists:

musicstream.local

5. Ensure that the required DNS records are created automatically.

Step 2: Install DHCP Role

1. Open **Server Manager**.
2. Click **Add Roles and Features**.
3. Select and install:

Dynamic Host Configuration Protocol (DHCP) Server

4. Complete the installation.

Step 3: Configure DHCP Scope

1. Open the **DHCP Console**.
2. Create a new scope.
3. Configure the scope as follows:
 - **IP Address Range:** 192.168.1.100 – 192.168.1.200
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.1.1
 - **DNS Server:** Server IP address
 - **Domain:** musicstream.local
4. Activate the scope.

Result

All client devices will:

- Get automatic IP addresses.
- Resolve domain names through DNS.
- Use the musicstream.local domain.

3. Create and Share the PlaylistShares Folder

Step 1: Create Folder

Create the following folder:

C:\PlaylistShares

Step 2: Share the Folder

1. Right-click the **PlaylistShares** folder.
2. Select **Properties**.
3. Open the **Sharing** tab.
4. Click **Advanced Sharing**.
5. Enable **Share this folder**.
6. Set the Share Name to:

PlaylistShares

7. Open **Permissions**.
8. Remove **Everyone**.
9. Add:
 - marketing_user → **Read/Write**
 - tech_user → **Read/Write**
10. Apply the settings.

Step 3: Configure NTFS Permissions

1. Open the **Security** tab.
2. Add the required users.
3. Give the following permissions:
 - marketing_user → **Modify**
 - tech_user → **Modify**
4. Apply the settings.

Step 4: Map Network Drive

On the client machines:

1. Open **File Explorer**.
2. Select **This PC**.
3. Click **Map Network Drive**.
4. Enter the path:

\\musicstream.local\\PlaylistShares

5. Click **Finish**.

The shared folder will now be available as a network drive for the authorized users.

4. Create a Group Policy Object (GPO)

Objective

Create a GPO that enforces password complexity and disables USB storage access for users except the admin_user account.

Step 1: Create GPO

1. Open **Group Policy Management**.
2. Create a new GPO named:

SecurityPolicy_MusicStream

Step 2: Configure Password Complexity

Go to:

Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies

Enable:

Password must meet complexity requirements

Set:

Minimum password length: 8 or more characters.

Step 3: Disable USB Storage

Go to:

Computer Configuration → Administrative Templates → System → Removable Storage Access

Enable:

All Removable Storage classes: Deny all access

Step 4: Exclude Admin Account

1. Configure **Security Filtering** for the GPO.
2. Remove admin_user from the applicable filtering.
3. Apply the GPO to the required users/computers except the admin_user account.

5. Restore a Deleted User Account from Backup

Step 1: Open Windows Server Backup

1. Open **Server Manager**.
2. Go to:

Tools → Windows Server Backup

Step 2: Start Recovery Wizard

1. Select **Recover**.
2. Choose the backup location.
3. Select a backup taken before the user was deleted.

Step 3: Restore Active Directory Data

Choose:

System State Recovery

Select:

Original location

Step 4: Restart in Directory Services Restore Mode (DSRM)

1. Restart the domain controller.
2. Enter **Directory Services Restore Mode (DSRM)**.
3. Log in using the DSRM credentials.

Step 5: Restore the Deleted User

1. Open **Active Directory Users and Computers**.
2. Navigate to:

musicstream.local → Users

3. Restore the deleted user from the backup, or recreate the user account if required by the recovery method.

Step 6: Verify the User

1. Check that the user can log in.
2. Verify the user's group memberships.
3. Verify that the required permissions and access are restored.

